

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2017
 Programme: BE Full Marks: 100
 Course: Microprocessor and Assembly Language Programming Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) You are required to buy a brand new personal computer. What would you want the specifications and features of the microprocessor in the system and why? Explain with reference to the evolution of microprocessors. 7
- b) Draw block diagram 8086 Microprocessor and explain its Execution Unit. 8
2. a) Write the function, addressing modes, size and name of machine cycles for the following instructions: 6
 - i. MOV A,B
 - ii. MVI A, 32H
 - iii. LXI H, 2030H
- b) Define Instruction cycle, Machine cycle and T-states. 3
- c) Draw the timing diagram for the 8085 instruction IN 34H. 6
3. a) Assuming: DS=1000H, BX=2030H and SI=2020H; State the addressing mode of the following 8086 instructions and find the physical address of the source location. 5
 - i. MOV AX, [1234H]
 - ii. MOV AX, [BX]
 - iii. MOV AX, [BX+SI]
 - iv. MOV AX, [BX+SI+5]
 - v. MOV AX, [SI+4]
- b) Write an 8085 ALP to subtract two 16-bit numbers and store the result in memory locations starting from 2017H. 5
- c) Explain various assembler directives in brief. 5
4. a) Write an assembly language program to find the sum of two no. which 7

is input by the user through the key board and display the sum in screen.

- b) You have given string data "Microprocessor and Assembly language Programming". Write an ALP to print "Microprocessor Programming" from the above given data. 8
5. a) How interrupt processing occurs in a microprocessor? Explain vector chain and polled interrupt. 8
- b) What do you mean by address decoding? Design an address decoding circuit to interface 4K×8 RAM, 8K×8 ROM and 16K×8 RAM with starting address 4000h. 7
6. a) Draw block diagram of 8254 PIT and explain in brief. 5
- b) How can we accommodate 20 interrupt sources with 8259 PIC? 5
- c) Explain the different control words of 8255A PPI. 5
7. Write short notes on: (Any two) 2×5
 - a) DMA controller
 - b) Macro Assembler
 - c) 8085 flag register